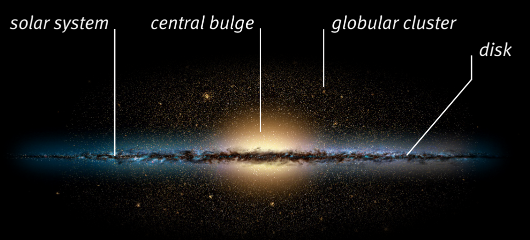
The Milky Way and beyond

The stars that surround Earth all belong to the Milky Way Galaxy, which is an enormous disk-shaped collection of stars, as well as gas and dust. The Sun and Earth are in a spiral arm of stars about two-thirds out from the center of the disk. It is an ideal position to observe the galaxy's stellar objects—its stars, clusters, and nebulae. We can also look beyond the Milky Way into a universe full of galaxies.

Fornax Cluster
Galaxies are not randomly spread though space; they exist in clusters huge distances apart that form superclusters. The Milky Way belongs to the Local Group, which, along with clusters such as the Fornax Cluster, forms the Local Supercluster.

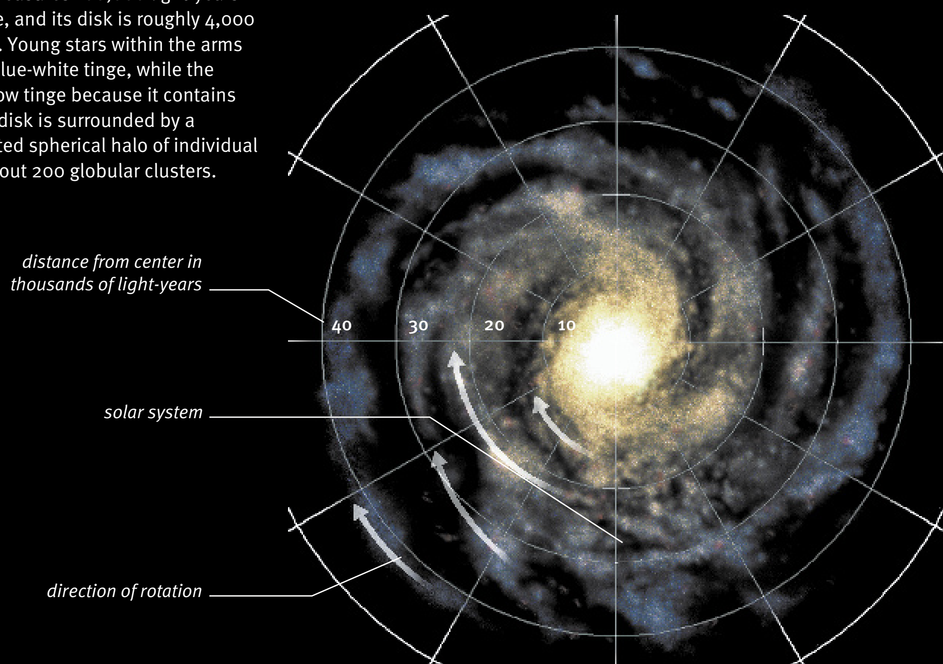


Edge-on view
The Milky Way measures 100,000 light-years from side to side, and its disk is roughly 4,000 light-years thick. Young stars within the arms give the disk a blue-white tinge, while the bulge has a yellow tinge because it contains older stars. The disk is surrounded by a sparsely populated spherical halo of individual old stars and about 200 globular clusters.

LOOKING AT THE MILKY WAY

The Milky Way is disk-shaped with a bulging center. Within the disk are spiral arms of stars. Stars also exist between the arms, but because the arm stars are young and bright, these are the ones that stand out. The center is packed with young and old stars. In total, there are about 500 billion stars.

From our position on Earth we can look into the disk and along its plane, either toward the center or away from it and out of the galaxy. The many stars in the disk appear as a river of milky light, hence the name Milky Way. The view to the center is seen in Sagittarius. A view through the plane and out of the galaxy is seen, for example, in the path above Orion's head. We see the other stars in our night sky (those not in the milky path) by looking perpendicular to the disk plane, up or down into the disk.



Bird's-eye view
This artist's impression shows what we think the Milky Way looks like from outside and above the disk. Gravity keeps the stellar objects in the Milky Way together. They do not travel as a solid disk but follow individual paths around the galaxy's center. They move at about the same speed; the closer to the center, the shorter the orbit time. The Sun's orbit takes 220 million years.

Milky path of light
When we look into the plane of the Milky Way's disk, we see its stars as a path of light. This path is broadest and brightest within the constellation of Sagittarius, but the view is dappled with dark clouds of dust that hide more distant stars. One dust patch is known as the Dark Horse (above, right of center). The horse's legs point to the right edge of the image and its head toward the top edge.